

Evgeny Redekop

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Research Interests

Quantum instrumentation bridging condensed matter and quantum information; Josephson microwave microscopy; nanoSQUID-on-tip probe; spatially resolved spectroscopy of correlated and topological quantum materials; cryogenic measurement and low-noise instrumentation.

Education

University of California, Santa Barbara

2021–2026 (expected)

Ph.D. in Physics

Advisor: Prof. Andrea F. Young

Dissertation: *Local magnetic imaging of correlated states in moiré systems using nanoSQUID-on-tip microscopy*

Skolkovo Institute of Science and Technology

2018–2020

M.Sc. in Materials Science

Lomonosov Moscow University

2014–2018

B.Sc. in Physics

Research Experience

Graduate Researcher, NSF Quantum Foundry

Jan 2021 – Present

University of California, Santa Barbara (Advisor: Andrea F. Young)

- Developed and optimized **nanoSQUID-on-tip (nSOT)** probes, improving magnetic flux sensitivity by over $50\times$ to $4\text{ n}\Phi_0/\sqrt{Hz}$, the highest reported to date (manuscript in preparation).
- Designed custom **UHV cryogenic deposition tools** and Python-based multi-instrument control software.
- Performed first **local magnetic imaging of fractional Chern insulators** in twisted MoTe_2 (*Nature*, 2024).
- Discovered **current-induced spin-orbit torque switching** in moiré Chern magnets (*Nature Physics*, 2023).
- Collaborated with theorists and experimentalists (L. Fu, A. MacDonald, L. Levitov, L. Glazman, K. Mak, J. Shan, X. Xu, S. Wilson, C. Palmstrøm).
- Supervised several undergraduate projects on cryogenic nSOT system development and fabrication yield optimization.

Scanning Electron Microscopy Superuser

Oct 2022 – May 2024

University of California, Santa Barbara

Trained and supported new users on Thermo Fisher Apreo C SEM system.

Student Researcher, Skolkovo Institute of Science and Technology

Oct 2018 – Oct 2020

Advisor: Prof. Albert G. Nasibulin

Built experimental setup for synthesis and characterization of metal-organic frameworks; conducted STM, FTIR, and THz spectroscopy measurements; developed data analysis pipelines in Python.

Awards and Fellowships

- **NSF Quantum Foundry Fellowship** (2021–2025) – Four-year competitive fellowship supporting quantum materials instrumentation.
- **Art of Science Competition – Honorable Mention** (2023), UCSB.
- **FASIE Grant for Young Entrepreneurs** (2015) – Foundation for Assistance to Small Innovative Enterprises.
- **Diploma with Distinction**, Skolkovo Institute of Science and Technology (2020).

Technical Skills

Cryogenic scanning probe microscopy; characterization tools (AFM, SEM); UHV thin-film deposition (In, Pb); Python/MATLAB for instrumentation control and data analysis; low-noise measurement system design; vacuum system assembly and debugging; nanofabrication and materials characterization.

Publications

- W. Li*, **E. Redekop***, C. Wang Beach*, C. Zhang*, X. Zhang, X. Liu, W. Holtzmann, C. Hu, E. Anderson, H. Park, T. Taniguchi, K. Watanabe, J.-H. Chu, L. Fu, T. Cao, D. Xiao, A. Young, X. Xu. *Universal Magnetic Phases in Twisted Bilayer MoTe_2* , arXiv:2507.22354 (2025)
- Caitlin L. Patterson, Owen I. Sheekey, Trevor B. Arp, Ludwig F. W. Holleis, Jin Ming Koh, Youngjoon Choi, Tian Xie, Siyuan Xu, Yi Guo, Hari Stoyanov, **Evgeny Redekop**, Canxun Zhang, Grigory Babikyan, David Gong, Haoxin Zhou, Xiang Cheng, Takashi Taniguchi, Kenji Watanabe, Martin E. Huber, Chenhao Jin, Étienne Lantagne-Hurtubise, Jason Alicea, Andrea F. Young, *Superconductivity and spin canting in spin-orbit-coupled trilayer graphene*, Nature, 641, 632–638 (2025)
- Ganesh Pokharel, Canxun Zhang, **Evgeny Redekop**, Brenden R Ortiz, Andrea N Salinas, Sarah Schwarz, Steven J Gomez Alvarado, Suchismita Sarker, Andrea F Young, Stephen D Wilson, *Evolution of charge correlations in the hole-doped kagome superconductor $\text{CsV}_{3-x}\text{Ti}_x\text{Sb}_5$* , Physics Review Materials (2025)
- **E. Redekop**, C. Zhang, H. Park, J. Cai, E. Anderson, O. Sheekey, T. Arp, G. Babikyan, S. Salters, K. Watanabe, T. Taniguchi, M. Huber, X. Xu, A. Young. *Direct magnetic imaging of fractional Chern insulators in twisted MoTe_2* , Nature 635, 584–589 (2024).
- Rajarshi Bhattacharyya, **Evgeny Redekop**, Mirko Bacani, *Phases of Chern insulator in moiré bilayers*, White paper, attocube systems AG (2024)
- Trevor Arp, Owen Sheekey, Haoxin Zhou, CL Tschirhart, Caitlin L Patterson, HM Yoo, Ludwig Holleis, **Evgeny Redekop**, Grigory Babikyan, Tian Xie, Jiewen Xiao, Yaar Vituri, Tobias Holder, Takashi Taniguchi, Kenji Watanabe, Martin E Huber, Erez Berg, Andrea F Young, *Intervalley coherence and intrinsic spin-orbit coupling in rhombohedral trilayer graphene*, Nature Physics, 20, 1413–1420 (2024)
- Wenjin Zhao, Kaifei Kang, Lizhong Li, Charles Tschirhart, **Evgeny Redekop**, Kenji Watanabe, Takashi Taniguchi, Andrea Young, Jie Shan, Kin Fai Mak, *Realization of the Haldane Chern insulator in a moiré lattice*, Nature Physics, 20, 275–280 (2024)
- C.L. Tschirhart*, **E. Redekop***, L. Li, T. Li, S. Jiang, T. Arp, O. Sheekey, T. Taniguchi, K. Watanabe, M. Huber, K. Mak, J. Shan, A. Young. *Intrinsic spin Hall torque in a moiré Chern magnet*, Nature Physics 19, 807–813 (2023).
- Victor I. Kleshch, Alexander A. Tonkikh, Sergey A. Malykhin, **Eugene V. Redekop**, Andrey S. Orekhov, Andrey L. Chuvilin, Elena D. Obratsova, and Alexander N. Obratsov, *Field emission from single-walled carbon nanotubes modified by annealing and CuCl doping*, Applied Physics Letters 109, 143112 (2016)

Mentorship

Supervised undergraduate researchers at UCSB on projects including nSOT fabrication optimization and cryothermal evaporator design (2021–2025). Alumni have continued to graduate programs at UCSB and UCSD.

Public Talks

- APS March Meeting 2025 – *Local imaging of ferromagnetism and topology in twisted bilayer MoTe_2*
- APS March Meeting 2024 – *Magnetic imaging of integer and fractional Chern insulators in $t\text{MoTe}_2$*
- APS March Meeting 2023 – *Intrinsic spin Hall torque in a moiré Chern magnet*